MULTI-NODE INFANT CRY DETECTION SYSTEM WITH WEARABLE CAREGIVER ALERT

Official Hardware Bill of Materials (BOM) & Component Costing Schedule

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1. Subsystem A: Cot Sensing Nodes (3 Dedicated Cot Units)

Item	Component Name	Description & Specifications	Hardware Pinout / Notes	Qty	Unit Price	Total Price
A1	ESP32-C3-DevKitC-02	RISC-V single-core @ 160MHz, 4MB Flash, 400KB SRAM, native 2.4GHz Wi-Fi/ESP-NOW	Node Core MCU (Cots 1, 2, 3)	3	6,000	18,000
A2	INMP441 Digital MEMS Mic	Omnidirectional 24-bit I2S digital microphone module, bottom-port acoustic inlet	SCK: GPIO5, WS: GPIO6, SD: GPIO7, L/R: GND	3	4000	12,000
A3	100kΩ Resistors (0.25W, 1%)	Metal film precision resistors for 2:1 battery monitoring voltage divider	GPIO 1 (ADC1_CH1) divider network	6		600
A4	100nF (0.1µF) Ceramic Caps	Multilayer ceramic capacitors (MLCC) for ADC noise filtering & microphone decoupling	1x across INMP441 VDD/GND, 1x on GPIO1	6		600
A5	3mm / 5mm Red LED	High-efficiency Red LED for battery status signaling (solid, slow blink, fast blink)	GPIO 4 via 330Ω resistor (Active-HIGH)	3		300
A6	3mm / 5mm Blue LED	High-efficiency Blue LED for ESP-NOW link status; pulls GPIO 9 HIGH at boot	3.3V -> 330Ω -> LED -> GPIO 9 (Active-LOW)	3		300
A7	330Ω Resistors (0.25W)	Current-limiting resistors for Red and Blue indicator LEDs	Red LED (GPIO 4) & Blue LED (GPIO 9)	6		600
A8	TP4056 1S Li-Ion Charger	1A constant-current/constant-voltage USB-C Li-Ion charging module with protection	Battery management (Cots 1, 2, 3)	3	1,500	4,500
A9	5V Step-Up Boost Converter	MT3608 micro DC-DC boost module (converts 3.7V Li-Po to 5.0V system rail)	Powers ESP32-C3 5V/VIN pin	3	2,000	6,000

A10	1S 3.7V 2000mAh Li-Po Cell	Rechargeable single-cell lithium-polymer battery with integrated PCM protection	Primary power source for cot nodes	3	2,000	6,000
A11	47µF - 100µF Electrolytic Caps	25V radial electrolytic buffer capacitors for RF transmit transient suppression	Placed across 5V and GND boost rails	3		1,000
A12	Solderless Breadboards (Half)	400-tie point solderless half-size breadboards for clean workbench layout	Subsystem prototype mounting	2	1,500	3,000

2. Subsystem B: Central Intelligence Hub (1 Unit)

Item	Component Name	Description & Specifications	Hardware Pinout / Notes	Qty	Unit Price	Total Price
B1	ESP32-S3-DevKitC-1 N16R8	Dual-Core Xtensa LX7 @ 240MHz, 16MB Quad-SPI Flash, 8MB Octal PSRAM	Central Edge-AI Hub (Strictly avoids GPIO 33-37)	1	25,000	25,000
B2	2.4" ILI9341 Color TFT Display	240x320 SPI Color TFT with onboard 3.3V LDO; real-time cot GUI dashboard	CS: 10, MOSI: 11, SCK: 12, DC: 13, RESET: 14	1	20,000	20,000
B3	100Ω Resistor (0.25W)	Current-limiting backlight ballast resistor for ILI9341 LED backlight	Placed between 3.3V rail and TFT LED pin	1	300	300
B4	TP4056 1S Li-Ion Charger	1A USB-C battery charging module with over-discharge and short protection	Central hub battery management	1	1,500	1,500
B5	5V Step-Up Boost Converter	MT3608 DC-DC step-up boost converter stepping 3.7V Li-Po up to 5.0V	Powers S3 5V/VIN and TFT VCC	1	2,000	2,000
B6	1S 3.7V 3000mAh Li-Ion / Li-Po	High-capacity rechargeable lithium cell for extended continuous hub runtime	Primary hub energy reservoir	1	2,000	2,000
B7	100µF 25V Electrolytic Cap	Radial decoupling capacitor across 5V boost converter output rail	Suppresses display and Wi-Fi current spikes	1	100	100
B8	100nF Ceramic Capacitor	High-frequency ceramic decoupling capacitor placed across ESP32-S3 3.3V rail	Power supply ripple rejection	1	100	100

B9	Solderless Breadboard (Full)	830-tie point solderless breadboard to mount ESP32-S3, TFT, and power modules	Central hub workbench mounting	1	1,500	1,500

3. Subsystem C: Caregiver Wearable Alert Wristband (1 Unit)

Item	Component Name	Description & Specifications	Hardware Pinout / Notes	Qty	Unit Price	Total Price
C1	ESP32-C3 SuperMini	Ultra-compact single-core RISC-V board with castellated pins and USB-C	Wearable MCU core (BLE Client)	1	6000	6000
C2	0.96" SSD1306 I2C OLED	128x64 monochrome graphic OLED display; displays cot ID & alarm banner	SDA: GPIO 4, SCL: GPIO 5, VCC: 3.3V	1	5,000	5,000
C3	IRLML2502TR (or AO3400A)	Logic-level N-Channel MOSFET in SOT-23 (Vth < 1.2V, Id > 4A, Rds(on) < 45mΩ)	Gate driven by GPIO 3 (switches motor GND)	1	1,000	1,000
C4	10mm Flat Coin ERM Motor	3V ~60mA eccentric rotating mass coin vibration motor for silent tactile alerts	Connected between 3.3V and MOSFET Drain	1	2,000	2,000
C5	1N5819 Schottky Diode	Fast 40V 1A Schottky flyback diode; protects MOSFET from inductive kickback	Cathode to 3.3V, Anode to MOSFET Drain	1	100	100
C6	100Ω Resistor (0.25W)	Gate series damping resistor to suppress high-frequency switching oscillations	Between GPIO 3 and MOSFET Gate	1	100	100
C7	10kΩ Resistor (0.25W)	Gate pull-down resistor to guarantee motor remains completely OFF at boot	Between MOSFET Gate and GND	1	100	100
C8	1S 3.7V 500mAh Li-Po Cell	Compact ultra-lightweight Li-Po pouch battery for wristband form factor	Wearable wristband power source	1	2,000	2,000
C9	TP4056 Micro/USB-C Board	Compact 1S Li-Po charging board with onboard battery protection	Wristband charging interface	1	1,500	1,500
C10	Mini Breadboard / Perfboard	170-tie point mini breadboard or compact prototyping perfboard	Wearable circuit prototyping	1	1,500	1,500

4. Subsystem D: Hardware Prototyping Accessories & Workbench Consumables

Item	Component Name	Description & Specifications	Hardware Pinout / Notes	Qty	Unit Price	Total Price
D1	Premium Breadboard Jumpers (M-M)	65-piece assorted male-to-male flexible solderless jumper wires (various lengths)	Inter-rail & IC breadboard connections	2 packs		2,400
D2	Breadboard Jumpers (M-F)	40-pin female-to-male Dupont jumper ribbon cables for displays & breakouts	ILI9341 TFT & INMP441 mic connections	2 packs		2,400

5. Grand Cost Summary Table

No.	Subsystem / Expenditure Category	Subtotal Cost (₦)
1	Subsystem A: 3× Cot Sensing Nodes (MCUs, INMP441 Mics, LEDs, Li-Po Batteries, Boosters)	₦52,900
2	Subsystem B: 1× Central Intelligence Hub (ESP32-S3 N16R8, ILI9341 Color TFT, 3000mAh Battery)	₦52,500
3	Subsystem C: 1× Caregiver Wearable (ESP32-C3 SuperMini, SSD1306 OLED, MOSFET, ERM Motor)	₦19,300
4	Subsystem D: Prototyping Accessories, Cables, Connectors & Solder Consumables	₦4,800
5	Contingency / Miscellaneous Workshop Cost (~10%)	₦12,950
TOTAL	**GRAND TOTAL ESTIMATED HARDWARE EXPENDITURE**	₦142,450